LAB EXPERIMENT 10

Experiment 10: Transformation-Based POS Tagging

Aim

To demonstrate transformation-based tagging by correcting initial POS tags using a simple transformation rule.

Algorithm

1. Start the program.
2. Define a sentence as a list of words.
3. Assign initial POS tags to the words.
4. Apply a transformation rule: if a word follows 'to' and is initially tagged as a noun, change its tag to VERB.
5. Display the final word-tag pairs.
6. Stop the program.

Code

```
words = ["I", "want", "to", "play", "football"]

# Initial tags assigned by a simple baseline tagger
tags = ["PRON", "VERB", "PART", "NOUN", "NOUN"]

# Transformation rule:
# A word following "to" should be tagged as a VERB
for i in range(1, len(words)):
    if words[i - 1].lower() == "to" and tags[i] == "NOUN":
        tags[i] = "VERB"

print("Word -> Final POS Tag")
for word, tag in zip(words, tags):
    print(word, "->", tag)
```

Output / Screenshot

```
words = [
    ("They", "NOUN"),
    ("play", "NOUN"),
    ("football", "NOUN")
]

# Transformation rule:
# Change "play" from NOUN to VERB

for i, (word, tag) in enumerate(words):
    if word.lower() == "play" and tag == "NOUN":
        words[i] = (word, "VERB")

print("Transformation-Based POS Tags:")

for word, tag in words:
    print(word, "->", tag)
```

```
Transformation-Based POS Tags:
They -> NOUN
play -> VERB
football -> NOUN
```

Result

Thus, transformation-based tagging was demonstrated by applying a rule to correct the initial tag of a word based on its context.